1 Gradient Descent

Define $g_t = \nabla f(x_t)$. The gradient descent is $x_{t+1} = x_t - \gamma g_t$, where γ is the step size. We assume f is differentiable anywhere.

Properties:

1. **Characterization of L-smoothness (equivalent)**: (3.3) $\frac{L}{2}x^\top x - f(x)$ is convex; (3.5) $\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\|$ for any x, y;
2. **Operations that perserve smoothness** (3.6): (i) Assume f_i are L_i-smooth and $\lambda_i > 0$, then $f := \sum_i \lambda_i f_i$ is $\sum_i \lambda_i L_i$-smooth. Proof by (3.3). (ii) Assume f is L-smooth and $g(x) = Ax + b$, then $f \circ g$ is $L\|A\|^2$-smooth.
3. **Characterization of μ-strongly convexity (equivalent)**: (3.11) $f(x) - \frac{\mu}{2}x^\top x$ is convex.

Analysis based on the first-order characterization:

1. Vanilla analysis (no further assumption): by first-order characterization, we can bound $f(x_t) - f(x^*) \le g_t^\top(x_t - x^*)$. The algorithm says $g_t = (x_t - x_{t+1})/\gamma$, thus $g_t^\top(x_t - x^*) = \frac{1}{\gamma}(x_t - x_{t+1})^\top(x_t - x^*)$. Therefore, $\sum_{t=0}^{T-1}(f(x_t) - f(x^*)) \le \sum_{t=0}^{T-1} g_t^\top(x_t - x^*) \le \frac{\gamma}{2}\sum_{t=0}^{T-1}\|g_t\|^2 + \frac{1}{2\gamma}\|x_0 - x^*\|^2$.
2. Lipschitz convex functions: bounded gradients $\|\nabla f(x)\| \le B$ for any x. (3.1) Assume $\|x_0 - x^*\| \le R$. Then the result of vanilla analysis says $\sum_{t=0}^{T-1}(f(x_t) - f(x^*)) \le \frac{\gamma}{2}B^2T + \frac{1}{2\gamma}R^2$. Choose $\gamma = \frac{R}{B\sqrt{T}}$ yields $\frac{1}{T}\sum_{t=0}^{T-1}(f(x_t) - f(x^*)) \le \frac{RB}{\sqrt{T}}$.

Analysis based on control over the quadratic term:

1. L-smooth functions (not requiring convexity): (3.7) with $\gamma := 1/L$, we have $f(x_{t+1}) \le f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2$. Proof: use smoothness definition and plug in $x_{t+1} - x_t = -\frac{1}{L}\nabla f(x_t)$.
2. Convex L-smooth functions: (3.8) with $\gamma := 1/L$, we have $f(x_T) - f(x^*) \le \frac{L}{2T}\|x_0 - x^*\|^2$. This means we only need $T \ge \frac{R^2L}{2\epsilon}$ to achieve error at most ϵ.
3. μ-strongly convex and L-smooth functions: (3.14) GD with $\gamma = 1/L$ yields $\|x_{t+1} - x^*\|^2 \le (1 - \frac{\mu}{L})\|x_t - x^*\|^2$ and $f(x_T) - f(x^*) \le \frac{L}{2}(1 - \frac{\mu}{L})^T\|x_0 - x^*\|^2$. This means we need $T \ge \frac{L}{\mu}\log\left(\frac{R^2L}{2\epsilon}\right)$ to achieve error at most ϵ. Proof: (i) replacing the first-order characterization in the vanilla analysis by the condition of μ-strongly convexity, we get $\|x_{t+1} - x^*\|^2 \le 2\gamma(f(x^*) - f(x_t)) + \gamma^2\|\nabla f(x_t)\|^2 + (1 - \mu\gamma)\|x_t - x^*\|^2$.

Optimizing without knowing L or B: The idea is to guess L and refine it gradually. The first guess is $L_0 := \frac{2\epsilon}{R^2}$. For each guess, we check if the sufficient decrease (3.7) holds. If (3.7) holds in the whole process $T = \frac{R^2L_i}{2\epsilon}$, then L_i is a successful guess, and we finish. If the guess is incorrect, we double $L_{i+1} = 2L_i$ and repeat.

Accelerated Gradient Descent

The AGD is to do $y_{t+1} = x_t - \frac{1}{L}\nabla f(x_t)$, $z_{t+1} = z_t - \frac{t+1}{2L}\nabla f(x_t)$ and $x_{t+1} = \frac{t+1}{t+3}y_{t+1} + \frac{2}{t+3}z_{t+1}$. Initialized with $y_0 = z_0 = x_0$.

2 Projected Gradient Descent

PGD: do constrainted gradient descent. $y_{t+1} = x_t - \gamma\nabla f(x_t)$ then $x_{t+1} = \Pi_X(y_{t+1}) = \mathbf{argmin}_X\|x - y_{t+1}\|^2$.

Analysis for PGD: the same result but proof adapted using projection inequalities.

1. Vanilla analysis: the same analysis for PGD gives $g_t^\top(x_t - x^*) = \frac{1}{2\gamma}\left(\gamma^2\|g_t\|^2 + \|x_t - x^*\|^2 - \|y_{t+1} - x^*\|^2\right)$. By projection inequality, we have $\|x_{t+1} - x^*\|^2 \le \|y_{t+1} - x^*\|^2$. Thus, the result of vanilla analysis is the same.
2. Lipschitz convex functions: (4.2) the same as GD since it only requires the result of vanilla analysis.
3. L-smooth functions: (4.3) with $\gamma = 1/L$, we have $f(x_{t+1}) \le f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2 + \frac{L}{2}\|y_{t+1} - x_{t+1}\|^2$. In addition, $f(x_{t+1}) \le f(x_t)$.
4. Convex L-smooth functions: (4.4) the same result as GD. Proof: we use a tighter inequality for vanilla analysis. Instead of using $\|x_{t+1} - x^*\|^2 \le \|y_{t+1} - x^*\|^2$, we now use $\|x_{t+1} - x^*\|^2 + \|y_{t+1} - x_{t+1}\|^2 \le \|y_{t+1} - x^*\|^2$, so the vanilla analysis results in $\sum_{t=0}^{T-1}(f(x_t) - f(x^*)) \le \frac{1}{2L}\sum_{t=0}^{T-1}\|g_t\|^2 + \frac{L}{2}\|x_0 - x^*\|^2 - \frac{L}{2}\sum_{t=0}^{T-1}\|y_{t+1} - x_{t+1}\|^2$. Combining this with (4.3) gets the result.
5. μ-strongly convex and L-smooth functions: (4.5) PGD with $\gamma = 1/L$ yields $\|x_{t+1} - x^*\|^2 \le (1 - \frac{\mu}{L})\|x_t - x^*\|^2$ and $f(x_T) - f(x^*) \le \frac{L}{2}(1 - \frac{\mu}{L})^T\|x_0 - x^*\|^2 + (1 - \frac{\mu}{L})^{T/2}\|\nabla f(x^*)\|\|x_0 - x^*\|$. This is still $O(\log(\frac{1}{\epsilon}))$ steps. Proof: μ-strongly convexity strengthens the vanilla analysis to be $g_t^\top(x_t - x^*) \le \frac{1}{2\gamma}(\gamma^2\|g_t\|^2 + \|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2 - \|y_{t+1} - x_{t+1}\|^2) - \frac{\mu}{2}\|x_t - x^*\|^2$. This makes the vanilla analysis to give $\|x_{t+1} - x^*\|^2 \le 2\gamma(f(x^*) - f(x_t)) + \gamma^2\|\nabla f(x_t)\|^2 + (1 - \mu\gamma)\|x_t - x^*\|^2 - \|y_{t+1} - x_{t+1}\|^2$.

3 PL Condition and Coordinate Descent

3.1 PL Condition

Definitions:

1. **Polyak-Lojasiewicz condition**: we say f satisfies PL condition if $\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu(f(x) - f(x^*))$.

Properties:

1. **Smooth functions satisfying PL condition can be solved by GD in $O(\log(\frac{1}{\epsilon}))$**: for such functions, we have $f(x_T) - f(x^*) \le (1 - \frac{\mu}{L})^T(f(x_0) - f(x^*))$.
2. **Strong convexity w.r.t. $\|\cdot\|_1$ and $\|\cdot\|_2$**: if f is μ-strongly convex w.r.t. $\|\cdot\|_1$, then f is also μ-strongly convex w.r.t. $\|\cdot\|_2$; if f is μ-strongly convex w.r.t. $\|\cdot\|_2$, then f is μ/d-strongly convex w.r.t. $\|\cdot\|_1$. Proof: use $\|x\|_1 \ge \|x\|_2$ and $\|x\|_2 \ge \frac{1}{\sqrt{d}}\|x\|_1$.
3. **Strong convexity w.r.t. $\|\cdot\|_1$ implies PL condition w.r.t. $\|\cdot\|_\infty$**: (5.9) if f has strong convexity w.r.t. $\|\cdot\|_1$, then $\frac{1}{2}\|\nabla f(x)\|_\infty^2 \ge \mu(f(x) - f(x^*))$. Proof: simply use $\nabla f(x)^\top(y - x) \le \|\nabla f(x)\|_\infty\|y - x\|_1$ in (5.2) instead.

3.2 Coordinate Descent

Analysis for coordinate-wise smooth functions satisfying PL condition:

1. **Randomized coordinate descent**: (5.6) choose coordinate uniformly at random and set $\gamma_i = 1/L$ where $L = \max_i L_i$, the randomized coordinate descent gets $\mathbb{E}(f(x_T) - f(x^*)) \le (1 - \frac{\mu}{dL})^T(f(x_0) - f(x^*))$.
2. **Importance sampling**: (5.7) choose coordinate i with probability $\frac{L_i}{\sum_j L_j}$ and define $\overline{L} = \frac{1}{d}\sum_{i=1}^d L_i$, we have $\mathbb{E}(f(x_T) - f(x^*)) \le (1 - \frac{\mu}{d\overline{L}})^T(f(x_0) - f(x^*))$.
3. **Steepest coordinate descent**: (5.8) choose coordinate $i = \mathbf{argmax}_i|\nabla_i f(x_t)|$, we have $f(x_T) - f(x^*) \le (1 - \frac{\mu}{dL})^T(f(x_0) - f(x^*))$. Proof: simply remove the expectation and use maximum is greater than average.
4. **Steepest descent for strong convexity w.r.t. $\|\cdot\|_1$**: (5.10) if f is μ_1-strongly convex w.r.t. $\|\cdot\|_1$, then with $\gamma_i = 1/L$, steepest descent gives $f(x_T) - f(x^*) \le (1 - \frac{\mu_1}{L})^T(f(x_0) - f(x^*))$.

4 Subgradient Methods

Hyperplane separation theorem: for two non-empty convex sets S and T, they can be separated by a hyperplane iff $\mathbf{rint}(S) \cap \mathbf{rint}(T) = \emptyset$.

Definitions:

1. **Subgradient**: let f be a convex function, then a vector g is a subgradient of f at x if $f(y) \ge f(x) + g^\top(y - x)$.
2. **Subdifferential**: the set of all subgradient at x is called the subdifferential of f at x, denoted as ∂f.
3. **Directional derivative**: the directional derivative of f at x along d is $f'(x; d) = \lim_{\delta \to 0^+}\frac{f(x + \delta d) - f(x)}{\delta}$.

Properties:

1. **subgradient = gradient, when convex and differentiable**: (6.2) if f is convex and differentiable at x, then $\partial f = \{\nabla f(x)\}$; (6.3) if f is only differentiable but not convex, then $\partial f \subseteq \{\nabla f(x)\}$. Proof: clearly $\{\nabla f(x)\} \subseteq \partial f$ given f convex; assume $\partial f \ne \emptyset$, let $y = x + \epsilon d$ for small ϵ, then $\frac{f(y) - f(x)}{\epsilon} \ge g^\top d$, which means $\nabla f(x)^\top d \ge g^\top d$ for any d. This implies $g = \nabla f(x)$.
2. **Bounded subgradient for convex functions = Lipschitz**: (6.5) if f is convex and $\mathbf{dom}(f)$ is open, then $\|g\| \le B$ for all x and $g \in \partial f(x)$ is equivalent to $|f(x) - f(y)| \le B\|x - y\|$ for all x, y. Proof: (i) one side: let $y = x + \epsilon g$ for some $\epsilon > 0$. Then $f(y) - f(x) \ge \epsilon\|g\|^2$. By B-Lipschitz, $f(y) - f(x) \le B\|y - x\| = \epsilon B\|g\|$. These two imply $\|g\| \le B$. (ii) the other side: for any x, y and $g \in \partial f(x)$, $f(x) - f(y) \le g^\top(x - y) \le \|g\|\|x - y\| \le B\|x - y\|$ and $f(y) - f(x) \ge g^\top(y - x) \ge -B\|y - x\|$.
3. **Convexity means subgradient almost everywhere**: (6.10) let f be convex and $x \in \mathbf{rint}(\mathbf{dom}(f))$, then $\partial f(x)$ is non-empty and bounded.
4. **Subgradient everywhere means convexity**: if $\mathbf{dom}(f)$ is convex and for any $x \in \mathbf{dom}(f)$, $\partial f(x)$ is non-empty, then f is convex.
5. **Subgradient and directional derivative**: (6.13) let f be convex and $x \in \mathrm{int}(\mathbf{dom}(f))$, then $f'(x; d) = \max_{g \in \partial f(x)} g^\top d$.

Calculus of subgradient:

1. For $h(x) = \lambda f(x) + \mu g(x)$ where $\lambda, \mu \geq 0$ and f, g both are convex, then $\partial h(x) = \lambda \partial f(x) + \mu \partial g(x)$.
2. For $h(x) = f(Ax + b)$ where f is convex, then $\partial h(x) = A^\top \partial f(Ax + b)$.
3. For $h(x) = \sup_{\alpha \in A} f_\alpha(x)$ and each $f_\alpha(x)$ is convex, then $\partial h(x) \supseteq \text{conv}\{\partial f_\alpha(x) : f_\alpha(x) = h(x)\}$.
4. For $h(x) = F(f_1(x), \ldots, f_m(x))$ where F is non-decreasing and convex, then $\partial h(x) \supseteq \{\sum_{i=1}^m d_i \partial f_i(x) : (d_1, \ldots, d_m) \in \partial F(y_1, \ldots, y_m)\}$.

4.1 Subgradient descent

Consider f convex (possibly non-differentiable) on a closed and convex set X. The subgradient descent does $x_{t+1} = \Pi_X(x_t - \gamma_t g(x_t))$, where $g(x_t) \in \partial f(x_t)$. When f is differentiable, this reduces to PGD. However, moving towards the negative direction of subgradient is not necessarily decreasing the objective. Therefore, we can only measure $\min_{t \leq T} f(x_t) - f^*$ instead of $f(x_T) - f^*$.

Analysis for convex functions:

1. **General analysis**: (6.17) starting from x_1, we have $\min_{1 \leq t \leq T} f(x_t) - f^* \leq \frac{1}{2}(\sum_{t=1}^T \gamma_t)^{-1}(\|x_1 - x^*\|^2 + \sum_{t=1}^T \gamma_t^2 \|g(x_t)\|^2)$ and $f(\hat{x}_T) - f^* \leq \frac{1}{2}(\sum_{t=1}^T \gamma_t)^{-1}(\|x_1 - x^*\|^2 + \sum_{t=1}^T \gamma_t^2 \|g(x_t)\|^2)$ for $\hat{x}_t = (\sum_{t=1}^T \gamma_t)^{-1}(\sum_{t=1}^T \gamma_t x_t)$. Proof: by definition,

$O(1/\sqrt{t})$ convergence under different stepsizes for convex Lipschitz functions:

1. $\gamma_t = \gamma$: $\epsilon_t = \frac{1}{2}(\frac{R^2}{T\gamma} + B^2\gamma) \to \frac{B^2}{2}\gamma$. Choose $\gamma_t = \frac{R}{B\sqrt{T}}$, we have $\epsilon_t \leq \frac{RB}{\sqrt{T}}$.
2. $\sum_t \gamma_t \to +\infty$ and $\gamma_t \to 0$: we have $\epsilon_t \to 0$. If set $\gamma_t = \frac{R}{B\sqrt{t}}$, we get $\epsilon_t = O(\frac{BR}{\sqrt{T}})$
3. $\sum_t \gamma_t \to +\infty$ and $\sum_t \gamma_t^2 < +\infty$: $\epsilon_t \to 0$.
4. Polyak stepsize $\gamma_t = \frac{f(x_t) - f^*}{\|g(x_t)\|^2}$: this makes the general analysis to give $\|x_{t+1} - x^*\|^2 \leq \|x_t - x^*\|^2 - 2\gamma_t g(x_t)^\top (x_t - x^*) + \gamma_t^2 \|g(x_t)\|^2 \leq \|x_t - x^*\|^2 - 2\gamma_t(f(x_t) - f^*) + \gamma_t^2 \|g(x_t)\|^2 = \|x_t - x^*\|^2 - \frac{(f(x_t) - f^*)^2}{\|g(x_t)\|^2} \leq \|x_t - x^*\|^2 - \frac{(f(x_t) - f^*)^2}{B^2}$, thus guarantees the decrease of $\|x_t - x^*\|$. This implies $\sum_t (f(x_t) - f^*)^2 \leq R^2 B^2$, thus $\epsilon_t = O(\frac{RB}{\sqrt{T}})$.

$O(1/t)$ convergence for strongly convex Lipschitz functions:

1. (6.18) Let f be μ-strongly convex. With $\gamma_t = \frac{1}{\mu t}$, we have $\min f(x_t) - f^* \leq \frac{B^2(\log(T)+1)}{2\mu T}$ and $f(\hat{x}_T) - f^* \leq \frac{B^2(\log(T)+1)}{2\mu T}$ for $\hat{x}_T = \frac{1}{T}\sum_t x_t$. Proof: by strong convexity, $f(y) - f(x) + g(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2$. Thus, general analysis gives $\|x_{t+1} - x^*\|^2 \leq \|x_t - x^*\|^2 - 2\gamma_t(f(x_t) - f^* + \frac{\mu}{2}\|x_t - x^*\|^2) + \gamma_t^2 \|g(x_t)\|^2$, which implies an upper bound on $f(x_t) - f^*$. Following the same steps in the general analysis, we get the desired result.
2. (6.19) Let f be μ-strongly convex. With $\gamma_t = \frac{2}{\mu(t+1)}$, we have $\min f(x_t) - f^* \leq \frac{2B^2}{\mu(T+1)}$ and $f(\hat{x}_T) - f^* \leq \frac{2B^2}{\mu(T+1)}$ for $\hat{x}_T = \sum_t \frac{2t}{T(T+1)} x_t$. Proof: the same analysis gives $t(f(x_t) - f^*) \leq \frac{\mu t(t-1)}{4}\|x_t - x^*\|^2 - \frac{\mu t(t+1)}{4}\|x_{t+1} - x^*\|^2 + \frac{B^2}{\mu(t+1)}$. The result follows.

Subgradient descent is asymptotically optimal for first-order subgradient methods: (Thm 6.20): for any $1 \leq t \leq n$ and $x_1 \in \mathbb{R}^n$, there exists a B-Lipschitz continuous function f and a convex set X with diameter R, s.t. for any first-order method that generates $x_t \in x_1 + \text{span}(g(x_1), \ldots, g(x_{t-1}))$, where $g(x_i) \in \partial f(x_i)$, we have $\min_{1 \leq s \leq t} f(x_s) - f^* \geq \frac{BR}{4(1+\sqrt{t})}$. In addition, there exists a B-Lipschitz continuous function f that is μ-strongly convex, s.t. $\min_{1 \leq s \leq t} f(x_s) - f^* \geq \frac{B^2}{8\mu t}$. Proof: W.l.o.g., assume $x_1 = 0$. Let $X = \{x : \|x\| \leq R/2\}$ and $f(x) = C \max_{1 \leq i \leq t} x_i + \frac{\mu}{2}\|x\|^2$ for some $C > 0, \mu > 0$. The subgradient of f is $\partial f(x) = \mu x + C \cdot \text{conv}\{e_i : i \text{ s.t. } x_i = \max_{1 \leq j \leq t} x_j\}$. The optima of f is $f^* = -\frac{C^2}{2\mu t}$. Consider $g(x) = Ce_i + \mu x$, where i is the first coordinate that $x_i = \max_{1 \leq j \leq t} x_j$. Since the algorithm runs for t iterations, the last dimension cannot be updated. Therefore, $\min_{1 \leq s \leq t} f(x_s) - f^* \geq \frac{C^2}{2\mu t}$. (1) Let $C = \frac{B\sqrt{t}}{1+\sqrt{t}}$ and $\mu = \frac{2B}{R(1+\sqrt{t})}$, we have $\max_{g \in \partial f(x)} \|g\| \leq C + \mu\|x\| \leq R$, and the first result follows. (2) Let $C = \frac{B}{2}$ and $\mu = \frac{B}{R}$, we have $\max_{g \in \partial f(x)} \|g\| \leq C + \mu\|x\| \leq R$, f is μ-strongly convex, and the second result follows.

4.2 Mirror Descent

Mirror descent only uses subgradients, and have the same aymptotic performance as subgradient descent. However, it may have better constants than subgradient descent. Subgradient descent is a special case of mirror descent.

Definitions:

1. **Bregman Divergence**: let $w(x)$ be strictly convex and continuously differentiable on a close convex X. $V_w(x, y) = w(x) - w(y) - \nabla w(y)^T (x - y)$
2. **Prox-mapping**: given an input x and vector ξ, the prox-mapping is defined as $\text{Prox}_x(\xi) = \textbf{argmin}_{u \in X}\{V_w(u, x) + \langle \xi, u \rangle\}$, where w is 1-strongly convex.
3. **Mirror Descent**: do $x_{t+1} = \text{Prox}_{x_t}(\gamma_t g(x_t)) = \textbf{argmin}_{x \in X}\{w(x) + \langle \gamma_t g(x_t) - \nabla w(x_t), x \rangle\}$. Recall that the subgradient descent is equivalent to $x_{t+1} = \textbf{argmin}_{x \in X}\{\frac{1}{2}\|x - x_t\|^2 + \langle \gamma_t g(x_t), x \rangle\}$ and $V_w(x, y) = \frac{1}{2}\|x - y\|^2$ for $w(x) = \frac{1}{2}\|x\|^2$. Therefore, subgradient descent is a special case of mirror descent.

Properties:

1. **Three point identity**: (6.26) $V_w(x, z) = V_w(x, y) + V_w(y, z) - \langle \nabla w(z) - \nabla w(y), x - y \rangle$. Proof: by definition.
2. **Generalized Pythagorean Theorem for Bregman Divergence**: (6.23) if $x^* = \textbf{argmin}_{x \in C} V_w(x, x_0)$ for some convex set C, then for any $y \in C$, we have $V_w(y, x_0) \geq V_w(y, x^*) + V_w(x^*, x_0)$. Proof: use $\nabla V_w(x, x_0)^\top |_{x=x^*} (y - x^*) \geq 0 \Leftrightarrow (\nabla w(x^*) - \nabla w(x_0))^\top (y - x^*) \geq 0$ and the three point identity.
3. **Convergence of mirror descent**: (6.28) let f be convex, we have $\min_{1 \leq t \leq T} f(x_t) - f^* \leq \frac{1}{\sum_{t=1}^T \gamma_t}(V_w(x^*, x_1) + \frac{1}{2}\sum_{t=1}^T \gamma_t^2 \|g(x_t)\|^2)$. Proof: since $x_{t+1} = \textbf{argmin}_{x \in X}\{w(x) + \langle \gamma_t g(x_t) - \nabla w(x_t), x \rangle\}$, by optimality we have $\langle \nabla w(x_{t+1}) + \gamma_t g(x_t) - \nabla w(x_t), x - x_{t+1} \rangle \geq 0$. Therefore, we have $\langle \gamma_t g(x_t), x_{t+1} - x \rangle \leq \langle \nabla w(x_{t+1}) - \nabla w(x_t), x - x_{t+1} \rangle = V_w(x, x_t) - V_w(x, x_{t+1}) - V_w(x_{t+1}, x_t)$. Using $\langle \gamma_t g(x_t), x_t - x_{t+1} \rangle \leq \frac{\gamma_t^2}{2}\|g(x_t)\|^2 + \frac{1}{2}\|x_t - x_{t+1}\|^2$ and noticing $V_w(x_{t+1}, x_t) \geq \frac{1}{2}\|x_t - x_{t+1}\|^2$ by 1-strongly convexity, we have $\langle \gamma_t g(x_t), x_t - x^* \rangle \leq V_w(x^*, x_t) - V_w(x^*, x_{t+1}) + \frac{\gamma_t^2}{2}\|g(x_t)\|^2$. The rest is the same as the proof of (6.17).

5 Stochastic Optimization

The goal is to $\min_{x \in X} F(x) = \frac{1}{n}\sum_{i=1}^n f_i(x)$, or more generally, $\min_{x \in X} F(x) = \mathbb{E}_\xi(f(x, \xi))$. For large n, computing the full gradient is expensive. For unknown $P(\xi)$, the gradient is intractable.

Stochastic gradient quality for smooth functions: (12.12) let $F(x) = \frac{1}{n}\sum_{i=1}^n f_i(x)$, where f_i is L_i-smooth and convex, and F has a global minimum x^*. Let $L_{\max} = \max_i\{L_i\}$, then for any x we have $\frac{1}{n}\sum_{i=1}^n \|\nabla f_i(x) - \nabla f_i(x^*)\|^2 \leq 2L_{\max}(F(x) - F(x^*))$. **Stochastic gradient descent**: do $x_{t+1} = \Pi_X(x_t - \gamma_t \nabla f(x_t, \xi_t))$. Thus, the *gradient is unbiased*: $\mathbb{E}_{i_t}(\nabla f_{i_t}(x_t) \mid x_t) = \sum_i \frac{1}{n}\nabla f_i(x_t) = \nabla F(x_t)$, or $\mathbb{E}(\nabla f(x_t, \xi_t) \mid \xi_{[t-1]}) = \nabla F(x_t)$. The step size should diminish, i.e., $\gamma_t \to 0$, to ensure convergence, as the stochastic gradient does not necessarily equal to zero at the optima.

Analysis:

1. **Strongly convex functions**: (12.3) assume $F(x)$ is μ-strongly convex and $\mathbb{E}(\|\nabla f(x, \xi)\|^2) \leq B^2$ for any $x \in X$. With $\gamma_t = \gamma/t$ for $\gamma > \frac{1}{2\mu}$, SGD satisfies $\mathbb{E}(\|x_t - x^*\|^2) \leq \frac{C(\gamma)}{t}$, where $C(\gamma) = \max\{\frac{\gamma^2 B^2}{2\mu\gamma - 1}, \|x_1 - x^*\|^2\}$. Proof: by projection inequality, $\|x_{t+1} - x^*\|^2 \leq \|x_t - \gamma_t \nabla f(x_t, \xi_t) - x^*\|^2 = \|x_t - x^*\|^2 - 2\gamma_t \langle \nabla f(x_t, \xi_t), x_t - x^* \rangle + \gamma_t^2 \|\nabla f(x_t, \xi_t)\|^2$. Taking expectation, we have $\mathbb{E}(\|x_{t+1} - x^*\|^2) \leq \mathbb{E}(\|x_t - x^*\|^2) - 2\gamma_t \mathbb{E}(\langle \nabla f(x_t, \xi_t), x_t - x^* \rangle) + \gamma_t^2 B^2$. Note that $\mathbb{E}(\langle \nabla f(x_t, \xi_t), x_t - x^* \rangle) = \mathbb{E}\Big[\mathbb{E}(\langle \nabla f(x_t, \xi_t), x_t - x^* \rangle \mid \xi_{[t-1]})\Big] = \mathbb{E}(\langle \nabla F(x_t), x_t - x^* \rangle)$. By strong convexity, $\langle \nabla F(x_t), x_t - x^* \rangle \geq \mu\|x_t - x^*\|^2$, thus $\mathbb{E}(\|x_{t+1} - x^*\|^2) \leq (1 - 2\mu\gamma_t)\mathbb{E}(\|x_t - x^*\|^2) + \gamma_t^2 B^2$. The result follows by induction.
2. **Convex functions**: (12.4) let F be convex and $\mathbb{E}(\|\nabla f(x, \xi)\|^2) \leq B^2$ for any $x \in X$. SGD satisfies $\mathbb{E}(F(\hat{x_T}) - F(x^*)) \leq \frac{R^2 + B^2 \sum_{t=1}^T \gamma_t^2}{2\sum_{t=1}^T \gamma_t}$ for $\hat{x_T} = \frac{\sum_{t=1}^T \gamma_t x_t}{\sum_{t=1}^T \gamma_t}$. Proof: use $\langle \nabla F(x_t), x_t - x^* \rangle \geq F(x_t) - F(x^*)$ in the proof of (12.3) gives $\gamma_t \mathbb{E}(F(x_t) - F(x^*)) \leq \frac{1}{2}\mathbb{E}(\|x_t - x^*\|^2) - \frac{1}{2}\mathbb{E}(\|x_{t+1} - x^*\|^2) + \frac{1}{2}\gamma_t^2 B^2$. The result follows by recursion and convexity.
3. **Strongly convex and smooth functions, constant step size**: (12.5) assume $F(x)$ is both μ-strongly convex and L-smooth, and $\mathbb{E}(\|\nabla f(x, \xi)\|^2) \leq \sigma^2 + c\|\nabla F(x)\|^2$. Then, with $\gamma_t = \gamma \leq \frac{1}{Lc}$, $\mathbb{E}(F(x_t) - F(x^*)) \leq \frac{\gamma L \sigma^2}{2\mu} + (1 - \gamma\mu)^{t-1}(F(x_1) - F(x^*))$.
4. **Non-convex but smooth function**: (12.8) assume $F(x) = \mathbb{E}(f(x, \xi))$ is L-smooth and $\mathbb{E}(\|\nabla f(x, \xi) - \nabla F(x)\|^2) \leq \sigma^2$, then with $\gamma_t = \min\{1/L, \frac{\gamma}{\sigma\sqrt{T}}\}$, SGD achieves $\mathbb{E}(\|\nabla F(\hat{x}_T)\|^2) \leq \frac{\sigma}{T}(2(F(x_1) - F(x^*))/\gamma + L\gamma)$, where $\hat{x}_T$ is selected uniformly at random from $\{x_1, \ldots, x_T\}$. Proof: by L-smoothness and

$x_{t+1} = x_t - \gamma_t \nabla f(x_t, \xi_t)$, we have $\mathbb{E}(F(x_{t+1}) - F(x_t)) \le \mathbb{E}(\nabla F(x_t)^\top (x_{t+1} - x_t) + \frac{L}{2}\|x_{t+1} - x_t\|^2) \le \mathbb{E}(-\gamma_t \nabla F(x_t)^\top \nabla f(x_t, \xi_t) + \frac{L\gamma_t^2}{2}\|\nabla f(x_t, \xi_t)\|)$. Using $\mathbb{E}(\nabla f(x_t, \xi_t) \mid x_t) = \nabla F(x_t)$ and $\mathbb{E}(\|\nabla f(x_t, \xi_t)\|^2 \mid x_t) \le \sigma^2 + \|\nabla F(x_t)\|^2$, this implies $\mathbb{E}(F(x_{t+1}) - F(x_t)) \le -\frac{\gamma_t}{2}\mathbb{E}(\|\nabla F(x_t)\|^2) + \frac{L\sigma^2\gamma_t^2}{2}$ since $\gamma_t \le 1/L$. The result follows by induction.

Variance Reduction Technique: the convergence guarantee of SGD with constant step size depends on the variance of the gradient. Reducing its variance makes it converge to a nearer point to the optimal.

1. **Mini-batch**: use a mini-batch to estimate gradient, i.e., $\nabla f(x_t, \xi_t) = \frac{1}{b}\sum_{i=1}^b \nabla f(x_t, \xi_{t,i})$.
2. **Momentum**: do $x_{t+1} = x_t - \gamma_t \hat{m}_t$, where $\hat{m}_t = c\sum_{\tau=1}^t \alpha^{t-\tau} \nabla f_{i_\tau}(x_\tau)$ is the weighted average of the past stochastic gradients.
3. **Control variate**: assume we want to estimate $\theta = \mathbb{E}X$, and we know a random variable Y that is highly correlated with X and $\mathbb{E}Y$ can be computed easily. Then $\hat{\theta}_\alpha := \alpha(X - Y) + \mathbb{E}Y$ has smaller bias and larger variance when α increases from 0 to 1. When $\alpha = 0$ it has zero variance and when $\alpha = 1$ it has zero bias.

Stochastic variance-reduced algorithms:

1. **Stochastic average gradient (SAG)**: use average of the past gradient as an estimate of the full gradient: $g_t = \frac{1}{n}\sum_{i=1}^n v_i^t$, where $v_i^t = \nabla f_{i_t}(x_t)$ for $i = i_t$ and $v_i^t = v_i^{t-1}$ otherwise. Equivalently, $g_t = g_{t-1} + \frac{1}{n}(\nabla f_{i_t}(x_t) - v_{i_t}^{t-1})$. This means SAG is as cheap as SGD. It has convergence rate linear in $\log 1/\epsilon$ for strongly convex and smooth functions.
2. **SAGA**: use $g_t = \nabla f_{i_t}(x_t) - v_{i_t}^{t-1} + \frac{1}{n}\sum_{i=1}^n v_i^{t-1}$ to make it unbiased. It has the same rate as SAG.
3. **Stochastic variance-reduced gradient (SVRG)**: use a fixed reference point to estimate the gradient: $g_t = \nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x}) + \nabla F(\tilde{x})$ and $x_{t+1} = x_t - \eta g_t$, where the reference point $\tilde{x}_t$ is updated only once a while. Idea: $\mathbb{E}(\|g_t - \nabla F(x_t)\|^2) \le \mathbb{E}(\|\nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x})\|^2) \le L_{\max}^2\|x_t - \tilde{x}\|^2$, so the variance is bounded by how close x_t is to $\tilde{x}$. Typical choice of $\tilde{x}$: each epoch s consists of m update, then use $\tilde{x}^s = \frac{1}{m}\sum_i x_t$ as the reference point of next epoch.

6 Frank-Wolfe Algorithm

This is to solve constrained optimization, similar to PGD.

Definitions:

1. **Linear minimization oracle (LMO)**: $\mathrm{LMO}_X(g) = \mathbf{argmin}_{z\in X}\, g^\top z$.
2. **Frank-Wolfe algorithm**: do $s = \mathrm{LMO}_X(\nabla f(x_t))$ and $x_{t+1} = (1-\gamma_t)x_t + \gamma_t s$. Benefits: (1) the iterates are always feasible for convex domain; (2) the algorithm is projection-free if we can solve LMO; (3) the iterates have a sparse representation, defined by the combination of LMO in the previous steps; (4) it is affine-invariant, meaning that affine equivalent problems can be solved at the same cost, i.e., if $g(x) = f(Ax + b)$ and $\mathbf{dom}(g) = A^{-1}(\mathbf{dom}(f) - b)$, then minimizing g is the same to minimizing f.
3. **Curvature constant**: the curvature constant of the constrained optimization is defined to be $C_{f,X} = \sup_{y=(1-\gamma)x+\gamma s, \gamma\in(0,1]} \frac{1}{\gamma^2}(f(y) - f(x) - \nabla f(x)^\top (y - x))$.

LASSO (L_1 domain): minimize $\|Ax - b\|^2$ subject to $\|x\|_1 \le 1$. Thus, $\mathrm{LMO}_X(g) = \mathbf{argmin}_{z=\pm e_i, i\in[n]}\, g^\top z = -\mathbf{sgn}(g_i)e_i$, where $i = \mathbf{argmax}|g_i|$. Therefore, we can compute LMO in $O(\log(d))$.

Duality gap as a certificate for optimization quality: We define the duality gap at x to be $g(x) = \nabla f(x)^\top (x - s)$, where $s = \mathrm{LMO}_X(\nabla f(x))$. (8.2) For convex f, $g(x) \ge f(x) - f(x^*)$. Proof: $g(x) = \nabla f(x)^\top x - \min_{z\in X} \nabla f(x)^\top z \ge \nabla f(x)^\top x - \nabla f(x)^\top x^* \ge f(x) - f(x^*)$.

Analysis:

1. **Decrease property**: (8.4) for $\gamma_t \in [0, 1]$, we have $f(x_{t+1}) \le f(x_t) - \gamma_t g(x_t) + \gamma_t^2 \frac{L}{2}\|s - x_t\|^2$, where $g(x_t)$ is the duality gap. Proof: by smoothness, $f(x_{t+1}) \le f(x_t) + \nabla f(x_t)^\top \gamma_t(s - x_t) + \gamma_t^2 \frac{L}{2}\|s - x_t\|^2 = f(x_t) - \gamma_t g(x_t) + \gamma_t^2 \frac{L}{2}\|s - x_t\|^2$.
2. **Affine-invariant decrease**: we have $f(x_{t+1}) \le f(x_t) - \gamma_t g(x_t) + \gamma_t^2 C_{f,X}$. Proof: plug in $x = x_t$, $y = (1-\gamma_t)x_t + \gamma_t s$ into the definition of curvature constant.
3. $\gamma_t = \frac{2}{t+2}$: (8.3) assume f is convex and L-smooth, and X is convex, closed and bounded, then $f(x_T) - f(x^*) \le \frac{2LR^2}{T+1}$, where $R = \max_{x,y\in X}\|x - y\|$. Proof: define $h(x) = f(x) - f(x^*)$, thus $h(x_{t+1}) \le h(x_t) - \gamma_t h(x_t) + \gamma_t^2 \frac{L}{2}\|s - x_t\|^2 \le (1-\gamma_t)h(x_t) + \gamma_t^2 C$, where $C = \frac{L}{2}R^2$. By induction, we have $h(x_t) \le \frac{4C}{t+1}$. (8.5) The same holds for $C_{f,X}$ instead of C.
4. **Convergence of duality gap**: (8.7) under the same setting, $g(x_t) \le O(\frac{C_{f,X}}{T+1})$.

7 Newton's Method

Find zeros: $x_{t+1} = x_t - \frac{f(x_t)}{f'(x_t)}$.

Find minimum: $x_{t+1} = x_t - \frac{f'(x_t)}{f''(x_t)}$. Generally, $x_{t+1} = x_t - \nabla^2 f(x_t)^{-1}\nabla f(x_t)$.

Analysis:

1. **Nondegenerate quadratic function**: (10.1) for $f(x) = \frac{1}{2}x^\top Mx - q^\top x + c$ with invertible symmetric M, Newton's method yields $x_1 = x^* = M^{-1}q$ for any x_0.
2. **Affine invariance**: (10.2) let f be twice differentiable, A invertible and $g(y) = Ay + b$. Define $N_h(x) = x - \nabla^2 h(x)^{-1}\nabla h(x)$, then $N_{f\circ g} = g^{-1} \circ N_f \circ g$.
3. **Bounded inverse Hessian and Lipschitz continuous Hessian**: (10.4) assume f is twice continuously differentiable, $\|\nabla^2 f(x)^{-1}\| \le 1/\mu$ and $\|\nabla^2 f(x) - \nabla^2 f(y)\| \le B\|x - y\|$ for any $x, y \in X$. Then we have $\|x_{t+1} - x^*\| \le \frac{B}{2\mu}\|x_t - x^*\|^2$ if a critical point x^* exists in X. Proof: Define $H(x) = \nabla^2 f(x)$. $x_{t+1} - x^* = x_t - x^* + H(x_t)^{-1}(\nabla f(x^*) - \nabla f(x_t)) = x_t - x^* + H(x_t)^{-1}\int_0^1 H(x_t + u(x^* - x_t))(x^* - x_t)du = H(x_t)^{-1}\int_0^1 (H(x_t + u(x^* - x_t)) - H(x_t))(x^* - x_t)du$.
4. **Fast convergence if near**: (10.5) under the assumption of (10.4), if $\|x_0 - x^*\| \le \mu/B$, then $\|x_T - x^*\| \le \frac{\mu}{B}(\frac{1}{2})^{2^T - 1}$. Proof: induction on (10.4).
5. **Global convergence for strongly convex and smooth functions**: with $\gamma = \mu/L$, (scaled) Newton's method $x_{t+1} = x_t - \gamma\nabla^2 f(x_t)^{-1}\nabla f(x_t)$ satisfies $f(x_t) - f^* \le (1 - \frac{\mu^2}{L^2})^t(f(x_0) - f^*)$. Note that the constant is worse than GD. Proof: expand $f(x_{t+1}) - f(x_t)$ by smoothness, then use $\frac{1}{L} \le \|H_t^{-1}\| \le \frac{1}{\mu}$ and $\|\nabla f(x_t)\|^2 \ge 2\mu(f(x_t) - f^*)$.

8 Quasi-Newton Methods

Computing and inverting the Hessian in Newton's method is costly. Quasi-Newton methods want to avoid this.

Secant condition: use approximation $f'(x_t) \approx \frac{f(x_t) - f(x_{t-1})}{x_t - x_{t-1}}$ to find zeros. Thus, the second method updates as: $x_{t+1} = x_t - f'(x_t)\frac{x_t - x_{t-1}}{f'(x_t) - f'(x_{t-1})}$. To generalize to higher dimensions, we want to find H_t such that $\nabla f(x_t) - \nabla f(x_{t-1}) = H_t(x_t - x_{t-1})$, and do $x_{t+1} = x_t - H_t^{-1}\nabla f(x_t)$.

Greenstadt's approach: update $H_t^{-1} = H_{t-1}^{-1} + E_t$ for some symmetric E_t with small $\|E_t\|_F^2 = \sum_{i,j} e_{i,j}^2$. To introduce more flexibility, we minimize $\|AE_tA^\top\|_F^2$ for some fixed invertible matrix A. The program is: minimize $\frac{1}{2}\|AE_tA^\top\|_F^2$ such that $E_t^\top = E_t$ and $E_t(\nabla f(x_t) - \nabla f(x_{t-1})) = x_t - x_{t-1} - H_{t-1}^{-1}(\nabla f(x_t) - \nabla f(x_{t-1}))$, which is the secant condition. Note that this is a convex program, so we solve by Lagrange method.

For simplicity, we write the program as: minimize $\frac{1}{2}\|AEA^\top\|_F^2$ such that $Ey = r$ and $E^\top - E = 0$. Define $f(E) = \frac{1}{2}\|AEA^\top\|_F^2$, thus $\nabla f(E) = A^\top AEA^\top A$. Note that the constraints are in fact linear over $e_{i,j}$. (11.3) Define $M = (A^\top A)^{-1}$ which is positive definite, then a solution E^* is optimal iff $E^* = M(\lambda y^\top + \Gamma^\top - \Gamma)M$, where $\lambda \in \mathbb{R}^{d\times 1}$ and $\Gamma \in \mathbb{R}^{d\times d}$. Solving the linear system gives $E^* = \frac{1}{y^\top My}(ry^\top M + Myr^\top - \frac{y^\top r}{y^\top My}Myy^\top M)$. This is called the Greenstadt method with parameter M.

BFGS:M = H'

L-BFGS: Define $\sigma = x_t - x_{t-1}$, $y = \nabla f(x_t) - \nabla f(x_{t-1})$, $H = H_{t-1}^{-1}$ and $H' = H_t^{-1}$, then BFGS can be written as $H' = (I - \frac{\sigma y^\top}{y^\top\sigma})H(I - \frac{y\sigma^\top}{y^\top\sigma}) + \frac{\sigma\sigma^\top}{y^\top\sigma}$.

9 Smoothing Techniques

9.1 Convex Conjugate

Convex conjugate (Legendre-Fenchel): for any f, its convex conjugate is given as $f^*(y) = \sup_{x\in\mathbf{dom}(f)}\{x^\top y - f(x)\}$.

Properties:

1. **Fenchel's inequality**: $f(x) + f^*(y) \ge x^\top y$ for any x, y. Proof by definition.
2. **Conjugate of a conjugate**: (7.2) If f is convex, lower semi-continuous and proper, then $(f^*)^* = f$. A proper convex function means $f(x) > -\infty$.
3. **Strong convexity implies nice conjugate**: (7.3) If f is μ-strongly convex, then f^* is continuously differentiable and $\frac{1}{\mu}$ smooth.

4. **Conjugate of function sum**: If f and g is lower semi-continuous and convex, then $(f+g)^*(x) = \inf_y\{f^*(y)+g^*(x-y)\}$.

9.2 Nesterov's Smoothing

Use $f_\mu(x) = \max_{y\in\mathbf{dom}(f^*)}\{x^\top y - f^*(y) - \mu d(y)\}$ as the surrogate function to minimize. $d(y)$ is a 1-strongly convex and nonnegative everywhere function, called the proximity function. Typical choices include $d(y) = \frac{1}{2}\|y-y_0\|^2$ and $d(y) = \frac{1}{2}\sum w_i(y_i - y_{0,i})^2$ for $w_i \geq 1$. This makes $f_\mu(x)$ to be $\frac{1}{\mu}$-smooth, as $f^*(y)+\mu d(y)$ is μ-strongly convex.

Approximation Error: for convex f with bounded $\mathbf{dom}(f^*)$, we have $f(x) - \mu D^2 \leq f_\mu(x) \leq f(x)$, where $D^2 = \max_{y\in\mathbf{dom}(f^*)} d(y)$. The tradeoff between approximation error and optimization error: $f(x) - f^* \leq [f(x) - f_\mu(x)] + [f_\mu(x) - \min_x f_\mu(x)]$. With AGD, we get $f(x_t) - f^* = O(\mu D^2 + \frac{R^2}{\mu t^2})$. To achieve approximation error ϵ, we need $\mu = O(\frac{\epsilon}{D^2})$. Therefore, $T_\epsilon = O(\frac{RD}{\epsilon})$.

9.3 Moreau-Yosida Smoothing

Use $f_\mu(x) = \min_y\{f(y) + \frac{1}{2\mu}\|x-y\|^2\}$. This is actually the special case of Nesterov's smoothing with $d(y) = \frac{1}{2}\|y\|^2$. Proof: $f_\mu^{Nes}(x) = \max_y\{x^\top y - f^*(y) - \frac{\mu}{2}\|y\|^2\} = (f^* + \frac{\mu}{2}\|\cdot\|^2)^*(x) = \inf_y\{f(y) + \frac{1}{2\mu}\|x-y\|^2\}$.

Properties: (1) $f_\mu(x)$ is $\frac{1}{\mu}$-smooth; (2) $\min_x f(x) = \min_x f_\mu(x)$. (3) GD reduces to proximal minimization: define $\mathbf{prox}_{\mu\cdot f} = \mathbf{argmin}_y\{f(y) + \frac{1}{2\mu}\|x-y\|^2\}$, then $x_{t+1} = x_t - \mu\nabla f_\mu(x_t) \Leftrightarrow x_{t+1} = \mathbf{prox}_{\mu\cdot f}(x_t)$.

Proximal operator: the proximal operator of convex function f at x is defined as $\mathbf{prox}_f(x) = \mathbf{argmin}_y\{f(y) + \frac{1}{2}\|x-y\|^2\}$. For continuous convex function f, $\mathbf{prox}_f(x)$ exists and is unique.

Properties:

1. **Fixed point**: If f is convex, then $x^* \in \mathbf{argmin}\, f(x) \Leftrightarrow x^* = \mathbf{prox}_f(x^*)$. Proof by definition.
2. **Non-expansive**: $\|\mathbf{prox}_f(x) - \mathbf{prox}_f(y)\| \leq \|x-y\|$. Proof: Let $\mu_x = \mathbf{prox}_f(x)$ and $\mu_y = \mathbf{prox}_f(y)$. By optimality condition, $x - \mu_x \in \partial f(\mu_x)$ and $y - \mu_y \in \partial f(\mu_y)$. By the monotonicity of gradient, $(x-\mu_x-(y-\mu_y))^\top(\mu_x-\mu_y) \geq 0$. This means $\|\mu_x-\mu_y\|^2 \leq (x-y)^\top(\mu_x-\mu_y) \leq \|x-y\|\|\mu_x-\mu_y\|$.
3. **Moreau decomposition**: For any x, $x = \mathbf{prox}_f(x) + \mathbf{prox}_{f^*}(x)$. Proof: use $u_x \in \partial f^*(x-u_x)$.

Proximal Point Algorithm: $x_{t+1} = \mathbf{prox}_{\gamma_t\cdot f}(x_t)$. (7.14) If f is convex, then $f(x_T) - f^* \leq \frac{\|x_0-x^*\|^2}{2\sum_{t=0}^{T-1}\gamma_t}$. Proof: by definition, $f(x_{t+1}) + \frac{1}{2\gamma_t}\|x_{t+1}-x_t\|^2 \leq f(x_t)$, which implies $f(x_{t+1}) - f(x_t) \leq -\frac{1}{2\gamma_t}\|x_{t+1}-x_t\|^2$. By optimality condition, $0 \in \partial f(x_{t+1}) + \frac{1}{\gamma_t}(x_{t+1}-x_t)$, which implies $\frac{x_t-x_{t+1}}{\gamma_t} \in \partial f(x_{t+1})$. Therefore, by the definition of subgradient, $f(x_{t+1}) - f^* \leq \frac{1}{\gamma_t}(x_t - x_{t+1})^\top(x_{t+1}-x^*) \leq \frac{1}{\gamma_t}[(x_t-x^*)^\top(x_{t+1}-x^*) - \|x_{t+1}-x^*\|^2]$. Using $(x_t-x^*)^\top(x_{t+1}-x^*) \leq \frac{1}{2}(\|x_t-x^*\|^2 + \|x_{t+1}-x^*\|^2)$, we get $f(x_{t+1}) - f^* \leq \frac{1}{2\gamma_t}[\|x_t-x^*\|^2 - \|x_{t+1}-x^*\|^2]$. The rest follows from non-increasing $f(x_t)$ and summing this inequality by t.

10 Min-Max Optimization

The min-max problem is defined as $\min_{x\in X}\max_{y\in Y}\phi(x,y)$.

Definitions:

1. **Saddle point and minimax point**: (x^*,y^*) is a global minimax point if $\phi(x^*,y) \leq \phi(x^*,y^*) \leq \max_{y'\in Y}\phi(x,y')$ for any x,y.
2. **Smoothness jointly in x and y**: $\|\nabla_x\phi(x_1,y_1) - \nabla_x\phi(x_2,y_2)\| \leq L(\|x_1-x_2\| + \|y_1-y_2\|)$ and $\|\nabla_y\phi(x_1,y_1) - \nabla_y\phi(x_2,y_2)\| \leq L(\|x_1-x_2\| + \|y_1-y_2\|)$.
3. **Duality gap**: defined to be $\max_y\phi(\hat{x},y) - \min_x\phi(x,\hat{y})$. an ϵ-saddle point.
4. **Monotone operators**: A operator F is monotone if $(F(u)-F(v))^\top(u-v) \geq 0$ for any u,v; it is μ-strongly monotone if $(F(u)-F(v))^\top(u-v) \geq \mu\|u-v\|^2$ for any u,v.

Properties:

1. **Minimax Theorem**: If X and Y are closed convex sets, one of them is bounded, and $\phi(x,y)$ is a continuous convex-concave function, then there exists a saddle point on $X\times Y$ and $\max_y\min_x\phi(x,y) = \min\max\phi(x,y)$.

Gradient Descent Ascent: do $x_{t+1} = \Pi_X(x_t - \gamma\nabla_x\phi(x_t,y_t))$ and $y_{t+1} = \Pi_Y(y_t + \gamma\nabla_y\phi(x_t,y_t))$. This simultaneously updates x and y.

Analysis:

1. **SC-SC and smooth fucntions**: (12.5) with $\gamma = \frac{\mu}{4L^2}$, GDA converges linearly: $\|x_{t+1}-x^*\|^2 + \|y_{t+1}-y^*\|^2 \leq (1-4\mu^2/L^2)(\|x_t-x^*\|^2 + \|y_t-y^*\|^2)$.
2. **C-C functions**: may not converge, e.g. $\phi(x,y) = xy$. Proof: $x_{t+1}^2 + y_{t+1}^2 = (x_t-\gamma y_t)^2 + (y_t+\gamma x_t)^2 = (1+\gamma^2)(x_t^2+y_t^2)$.

Extragradient Method:

$$\begin{aligned}
\mathbf{x}_{t+\frac{1}{2}} &= \Pi_{\mathcal{X}}\left(\mathbf{x}_t - \eta\nabla_{\mathbf{x}}\phi(\mathbf{x}_t,\mathbf{y}_t)\right),\\
\mathbf{y}_{t+\frac{1}{2}} &= \Pi_{\mathcal{Y}}\left(\mathbf{y}_t + \eta\nabla_{\mathbf{y}}\phi(\mathbf{x}_t,\mathbf{y}_t)\right)\\
\mathbf{x}_{t+1} &= \Pi_{\mathcal{X}}\left(\mathbf{x}_t - \eta\nabla_{\mathbf{x}}\phi\left(\mathbf{x}_{t+\frac{1}{2}},\mathbf{y}_{t+\frac{1}{2}}\right)\right)\\
\mathbf{y}_{t+1} &= \Pi_{\mathcal{Y}}\left(\mathbf{y}_t + \eta\nabla_{\mathbf{y}}\phi\left(\mathbf{x}_{t+\frac{1}{2}},\mathbf{y}_{t+\frac{1}{2}}\right)\right)
\end{aligned}$$

This is different from two steps of GDA, because the second step still uses the value of t. Essentially, EG is to use better gradients than GDA. EG has $O(1/T)$ convergece rate of the duality gap for averaged $x_{t+1/2}$ and $y_{t+1/2}$ in the C-C setting with bounded domain, and $O(\log(\frac{1}{\epsilon}))$ complexity in the SC-SC setting, which is optimal.

Optimistic GDA

$$\begin{aligned}
\mathbf{x}_{t+1} &= \mathbf{x}_t - 2\eta\nabla_{\mathbf{x}}\phi(\mathbf{x}_t,\mathbf{y}_t) + \eta\nabla_{\mathbf{x}}\phi(\mathbf{x}_{t-1},\mathbf{y}_{t-1})\\
\mathbf{y}_{t+1} &= \mathbf{y}_t - 2\eta\nabla_{\mathbf{y}}\phi(\mathbf{x}_t,\mathbf{y}_t) + \eta\nabla_{\mathbf{y}}\phi(\mathbf{x}_{t-1},\mathbf{y}_{t-1})
\end{aligned}$$

This is equivalent to:

$$\begin{aligned}
\mathbf{x}_{t+\frac{1}{2}} &= \Pi_{\mathcal{X}}\left(\mathbf{x}_t - \eta\nabla_{\mathbf{x}}\phi(\mathbf{x}_{t-\frac{1}{2}},\mathbf{y}_{t-\frac{1}{2}})\right), & \mathbf{y}_{t+\frac{1}{2}} &= \Pi_{\mathcal{Y}}\left(\mathbf{y}_t + \eta\nabla_{\mathbf{y}}\phi(\mathbf{x}_{t-\frac{1}{2}},\mathbf{y}_{t-\frac{1}{2}})\right)\\
\mathbf{x}_{t+1} &= \Pi_{\mathcal{X}}\left(\mathbf{x}_t - \eta\nabla_{\mathbf{x}}\phi\left(\mathbf{x}_{t+\frac{1}{2}},\mathbf{y}_{t+\frac{1}{2}}\right)\right), & \mathbf{y}_{t+1} &= \Pi_{\mathcal{Y}}\left(\mathbf{y}_t + \eta\nabla_{\mathbf{y}}\phi\left(\mathbf{x}_{t+\frac{1}{2}},\mathbf{y}_{t+\frac{1}{2}}\right)\right)
\end{aligned}$$

OGDA has similar convergence rate as EG in the SC-SC and C-C settings.

Proximal Point Algorithm

do $(x_{t+1},y_{t+1}) = \mathbf{argmin}_x\mathbf{argmax}_y\{\phi(x,y) + \frac{1}{2\eta}\|x-x_t\|^2 - \frac{1}{2\eta}\|y-y_t\|^2\}$.

This is equivalent to

$$\begin{aligned}
\mathbf{x}_{t+1} &= \Pi_{\mathcal{X}}\left(\mathbf{x}_t - \eta\nabla_{\mathbf{x}}\phi\left(\mathbf{x}_{t+1},\mathbf{y}_{t+1}\right)\right)\\
\mathbf{y}_{t+1} &= \Pi_{\mathcal{Y}}\left(\mathbf{y}_t + \eta\nabla_{\mathbf{y}}\phi\left(\mathbf{x}_{t+1},\mathbf{y}_{t+1}\right)\right)
\end{aligned}$$

PPA converges with $O(1/T)$ in the C-C setting.

Nash equilibrium for concave games: $u_i(x_i^*,x_{-i}^*) \geq u_i(x_i,x_{-i}^*)$ for any $x_i \in X_i$. First-order characterization: in the concave game, Nash equilibria can be characterized as $\nabla_i u_i(x_i^*,x_{-i}^*)^\top(x_i - x_i^*) \leq 0$.